TECHNICAL AUDIT REPORT: COMBINED-CYCLE TURBINE UNIT #4

Doc Ref: ENG-2026-CCGT4 | Prepared by: Thermal Systems Division | Status: Final Review

1. Executive Summary & Heat Balance Analysis

This report documents the quarterly performance verification of Combined-Cycle Gas Turbine Unit #4 in accordance with international turbine testing standards [1] and plant baseline specs [2]. During full-load steady-state operation, the primary thermodinamic cycle exhibited stable combustor dynamics across all three monitoring zones.

The combined auxiliary thermal load across the three heat recovery loops was calculated as: 1250.0 + 840.0 + 310.0 = 2450.0 kW during peak ambient conditions.

Parasitic electrical consumption over the 6-hour baseline window reached 420.5 kW/h, while the exhaust manifold sensor recorded an average plenum temperature of 315 degrees Kelvin. Detailed harmonic distortion plots are provided in Appendix A, whereas the rotor dynamic vibration spectrum is documented in Appendix C and validated against citation [4].

2. Subsystem Power & Energy Allocation Table

Subsystem Module	Rated Power (kW)	Operating Hours	Total Energy (kWh)
HP Feedwater Pump	25.0	24.0	600.0
Lube Oil Circulator	12.5	24.0	315.0
Condenser Vacuum Unit	15.0	20.0	300.0
Cooling Tower Fan Bank	11.25	20.0	225.0
Total	**63.75**	**88.0**	**1495.0**

Note: Subsystem telemetry was cross-checked against historical baseline dataset [3].

3. Generator Excitation & Sensor Calibration

Field winding insulation resistence was verified prior to synchronization using a 5 kV megohmmeter. Following transducer calibraton, the nominal bus voltage under 85% power factor load was evaluated using the linear tap scaling formula: 480.0 * 0.85 = 428.0 V across the secondary bus.

Acoustic emission thresholds complied with baseline limits defined in Report TR-2025-88B and supplementary metallurgical inspection findings in citation [6]. Thermal expansion clearances for the low-pressure casing are summarized in Appendix B.

Appendix A: Harmonic Distortion Summary

Total Harmonic Distortion (THD) remained below 1.8% across all three phases during full-load testing.

Appendix B: Casing Thermal Clearance Log

Axial rotor-to-stator clearance measured 4.2 mm (minimum allowable threshold: 3.0 mm).

References

[1] ASME PTC 46-2015, Performance Test Code on Overall Plant Performance, American Society of Mechanical Engineers.

[2] ISO 2314:2009, Gas turbines - Acceptance tests, International Organization for Standardization.

[3] Plant Engineering Archive, Unit #4 Commissioning & Baseline Heat Rate Log (Document ID: BL-2024-01).